DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& also for CS



Calculus and Optimization

Functions and Graphs

Lecture No. 01



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Topics to be Covered



Topic

Functions & Basic Graphs

Function: special type of Relationship b/n x & y s.t this is following certain conditions (Domain(x)) i.e. $y = f(x)$

1. Types of func^n → Algebraic func^n, Transcedental func^n
2. Types of func^n → Even func^n, odd func^n, Neither Even Nor odd.
3. Types of func^n → Periodic func^n, Non Periodic func^n

INFORMATION

Types of functions

- ALGEBRAIC function
  - (1) Polynomial func^n
  - (2) Rational func^n
  - (3) Irrational func^n
  - (4) Piecewise func^n
    - Mod func^n
    - Signum func^n
    - G.I.f.
    - L.I.f
    - F.P.f
- TRANSCEDENTAL function
  - (1) Exponential func^n
  - (2) log function.
  - (3) Trigonometric func^n
  - (4) Inverse Trig. functions.

G.I.f = Greatest Integer Func^n (Floor func^n)
L.I.f = Least Integer func^n (Ceiling func^n)
F.P.F = Fractional Part Func^n

PIECEWISE func^n → If func^n is defined by Multiple sub function

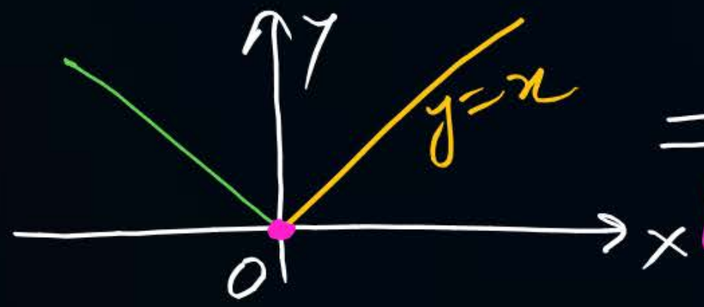
s.t, Domain of each subfunction is different

then function is called Piecewise func^n

for eg. Mod func^n, Signum func^n, G.I.F, L.I.F, Fractional Part func^n etc

MODULUS func^n →

$$y=|x|=\begin{cases}-x, & x<0\\ +x, & x\geq 0\end{cases}$$



$$=\boxed{\max\{x,-x\}}$$

eg $|5|=+5$, $|-5|=-(-5)=+5$

eg $|5|=\max\{5,-5\}=5$

$|-5|=\max\{-5,-(-5)\}$

$=\max\{-5,+5\}=+5$

special Def^n of $|x|$ → $y=\boxed{|x|=\sqrt{x^2}}$

eg $|5|=\sqrt{(5)^2}=\sqrt{25}=+5$

& $|-5|=\sqrt{(-5)^2}=\sqrt{25}=+5$

Q if $x^2-25=0\Rightarrow x=?$

(M-I) $(x-5)(x+5)=0\Rightarrow x=-5$ or 5

(M-II) $x^2-25=0\Rightarrow x=\sqrt{25}=\pm 5$ WRONG Explanation.

(M-III) $x^2-25=0\Rightarrow x^2=25$ ↓ Correct Version.

$x=\pm(\sqrt{25})$

$=\pm(+5)$

$x=\pm 5$

eg $[(-9)^2]^{\frac{1}{2}} = ?$

$= [(-9)^2]^{\frac{1}{2}} = -9$ ✗

$= \sqrt{(-9)^2} = \sqrt{81} = +9$ ✓

$\therefore \sqrt{(-9)^2} = \sqrt{81} = |9| = +9$

eg: the Number of solutions $x^2 - 5|x| + 6 = 0$ is/are ?

Case I: if $x < 0$ then

$x^2 - 5|x| + 6 = 0$

$x^2 - 5(-x) + 6 = 0$

$x^2 + 5x + 6 = 0$

$(x+2)(x+3) = 0$

$x = -2, -3$

Case II: if $x \geq 0$ then

$x^2 - 5|x| + 6 = 0$

$x^2 - 5(+x) + 6 = 0$

$x^2 - 5x + 6 = 0$

$(x-2)(x-3) = 0$

$x = 2 \& 3$

i.e. given eqn has four sol^{n}s.

Q. The No. of Sol^{n}s of $x^2+5|x|+6=0$ is/are ?

(a) 0

(b) 2

(c) 4

(d) ∞

M-I

Case I: if $(x<0)$ then

$x^2+5|x|+6=0$

$x^2+5(-x)+6=0$

$x^2-5x+6=0$

$(x-2)(x-3)=0$

$x=2$ & 3

(WRONG $\because x<0$)

Case II: if $(x>0)$ then

$x^2+5|x|+6=0$

$x^2+5(+x)+6=0$

$x^2+5x+6=0$

$(x+2)(x+3)=0$

$x=-2$ & -3

(WRONG $\because x>0$)

M-II (Shortcut) $\rightarrow$ $x^2+5|x|+6=0$

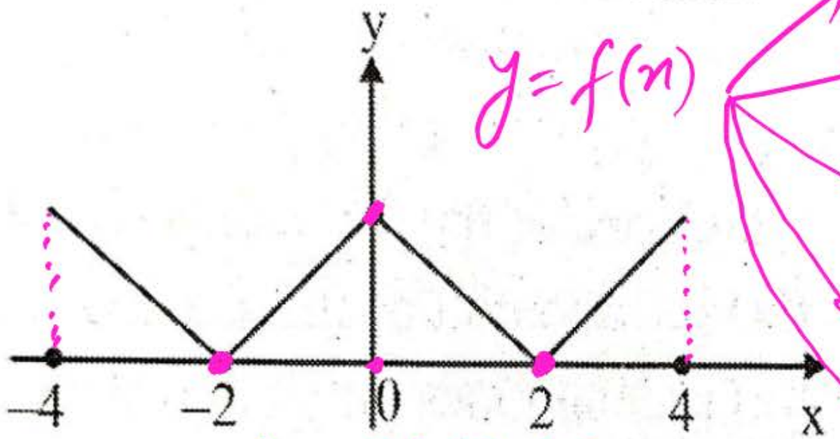
(+ve No.) + (+ve No.) + (+ve No.) $=0$ $\Longrightarrow$ Contradiction

So given eq^{n} has No Sol^{n}.

[MCQ]

[GATE-ME-2023: 1M]

The figure shows the plot of a function over the interval [–4, 4], which one of the options given CORRECTLY identifies the function?



$y = f(x)$

At $x = 2$, $y = 0$

At $x = -2$, $y = 0$

At $x = 4$, $y =$ +ve

At $x = -4$, $y =$ +ve

At $x = 0$, $y =$ +ve

(a) $y = |2 - x|$ (F) $\because y(-2) \neq 0$

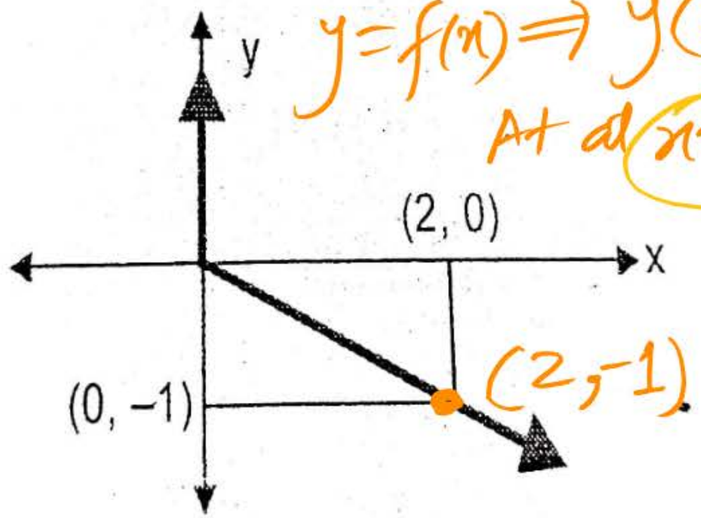
(b) $y = |2 - |x||$

(c) $y = |2 + |x||$ (F) $\because y(-2) \neq 0$

(d) $y = 2 - |x|$ (F) $\because y(4) =$ -ve (WRONG)

Ans (b) ✓

Choose the most appropriate equation for t
function drawn as a thick line, in the piot belo



$y = f(x) \Rightarrow y(2) = -1$

At at $x = 2, y = -1$

(a) $x = y - |y|$ (b) $x = -(y - |y|)$

(c) $x = y + |y|$ (d) $x = -(y + |y|)$

[GATE-2015-CS-SET-3; 2 Ma

(a) $x = y - |y| \Rightarrow$ At $y = -1$

$x = -1 - |-1| = -1 - 1 = -2$

So (a) is wrong.

(c) $x = y + |y| \Rightarrow$ At $y = -1$

$x = -1 + |-1| = -1 + 1 = 0$

So (c) is also wrong.

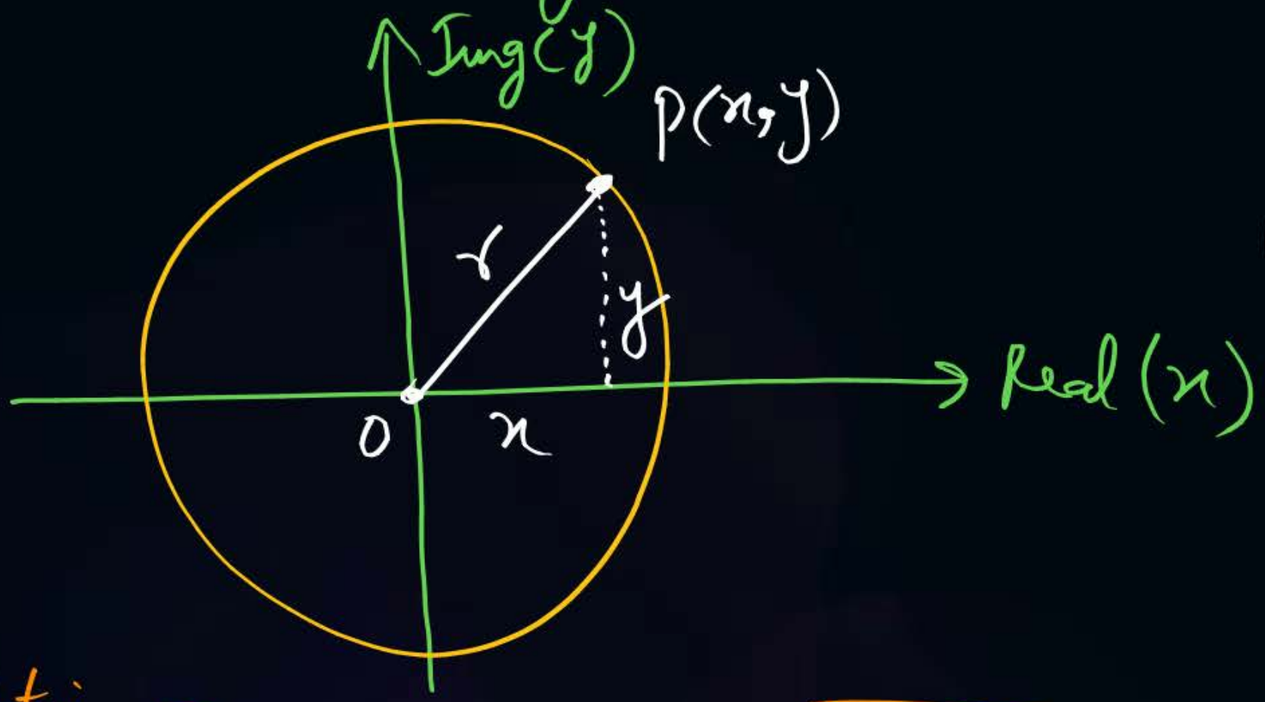
(b) $x = -(y - |y|)$, At $y = -1$

$x = -(-1 - |-1|) = -(-1 - 1) = 2$

So (b) is Correct.

Note: $y=(|x|=\sqrt{x^2}) \Rightarrow$ Sq. Root of Real Number is always +ve.

w.k.t $z=x+iy \simeq (x,y)$ = Point



$z=x+iy$ then $r^2=x^2+y^2$

$r=OP=|z|=\sqrt{x^2+y^2}$

For Real 'z', Put $y=0$

$\Rightarrow z=x+i(0)=x$

& $|z|=\sqrt{x^2+(0)^2}$

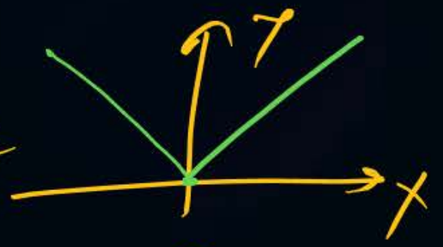
$|x|=\sqrt{x^2}$ = +ve Number.

Hence Proved

MAG

Note: IOTA :- $i=\sqrt{-1}$ & $i^2=-1$

⊛ $y = |x| =$



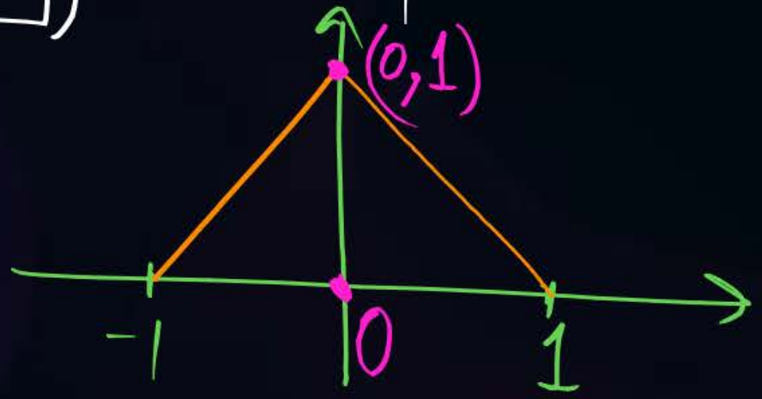
⊛ $y = -|x|$ =



⊛ Draw the Graph of $y = \begin{cases} 1+x, & -1 \le x < 0 \\ 1-x, & 0 \le x \le 1 \end{cases}$

Sol: $\boxed{y = 1-|x|}$, $-1 \le x \le 1$

$y = -|x| + 1$



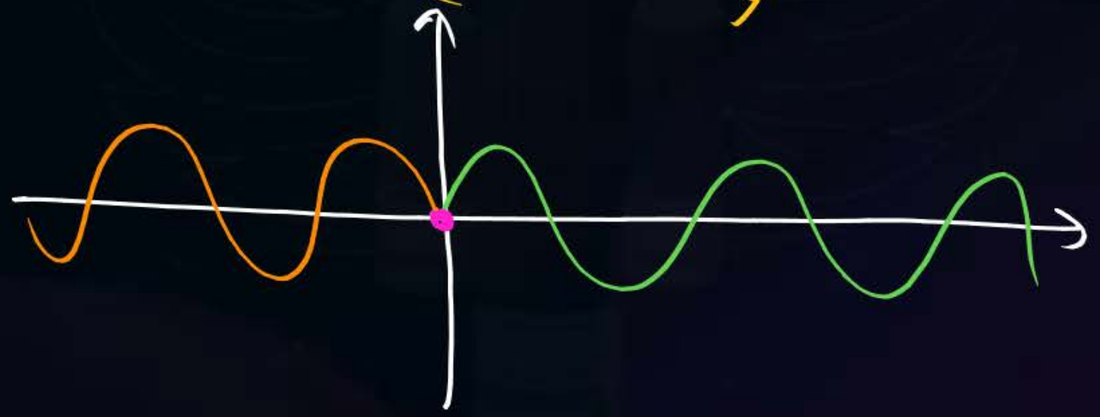
⊛ $y = \sin x$



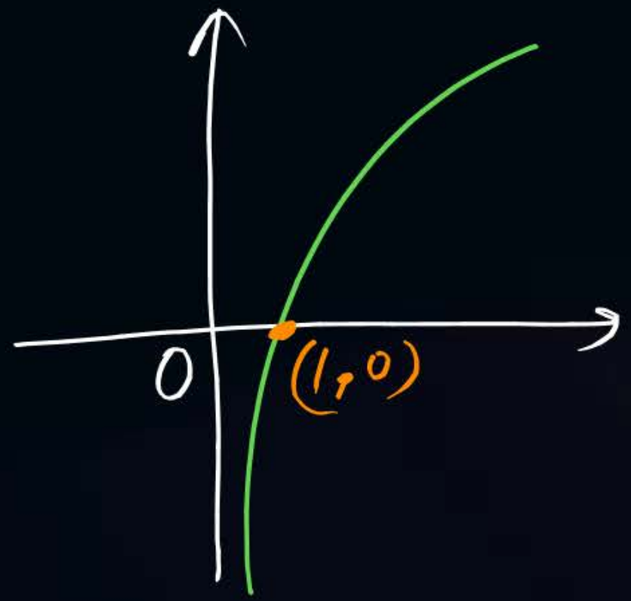
⊛ $y = |\sin x|$



⊛ $y = \sin|x| = \begin{cases} -\sin x, & x < 0 \\ \sin x, & x > 0 \end{cases}$



(*) $y = \log x$



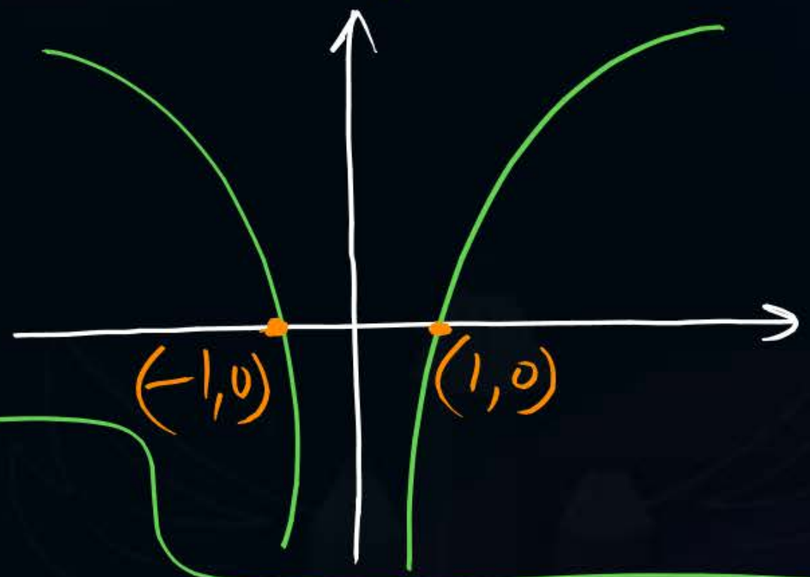
Domain $(x) = (0, \infty)$

(*) $y = |\log x|$

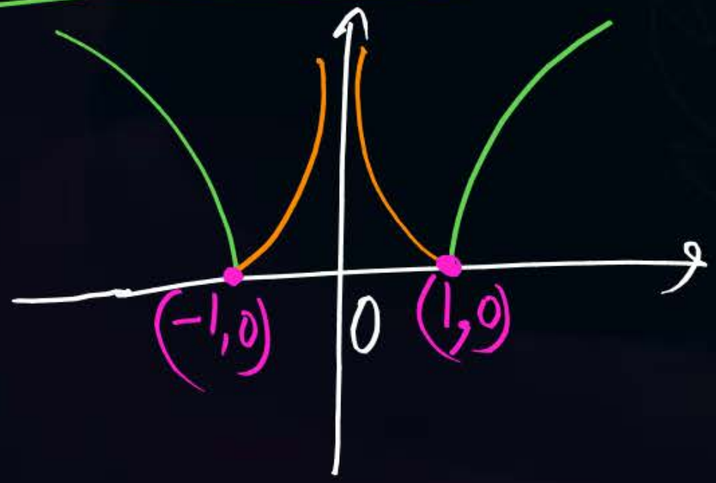


(*) $y = \log|x| = \begin{cases} \log(-x), & x < 0 \\ \log(x), & x > 0 \end{cases}$

is Domain $(x) = (-\infty, \infty)$

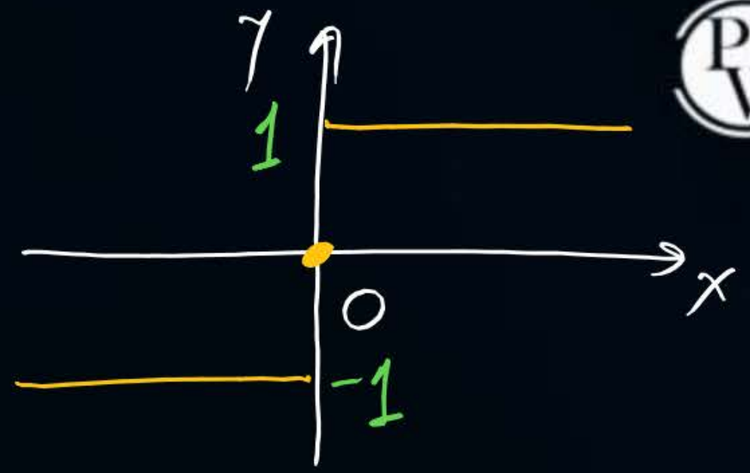


(*) $y = |\log|x||$ $\Rightarrow$



⊛ Signum func^n →

$$y = sgn(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \\ 0, & x = 0 \end{cases}$$



eg $sgn(0) = 0$, $sgn(1.3) = 1$, $sgn(1.9) = 1$, $sgn(2.7) = 1$ ----

& $sgn(-3.4) = -1$, $sgn(-0.2) = -1$, $sgn(-4.5) = -1$ ----

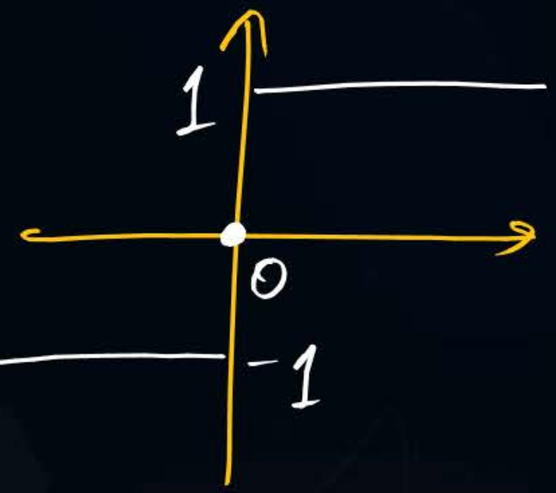
$sgn(1) = 1$, $sgn(2) = 1$, $sgn(3) = 1$ ------

$sgn(-1) = -1$, $sgn(-2) = -1$, $sgn(-3) = -1$

PODCAST (Analysis) →

① $y = \frac{|x|}{x} = \frac{x}{x}\left(\frac{|x|}{|x|}\right) = \frac{|x|^2}{x|x|} = \frac{x^2}{x|x|} = \frac{x}{|x|}$, $x \neq 0$.

i.e. $y = \text{sgn}(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases} =$

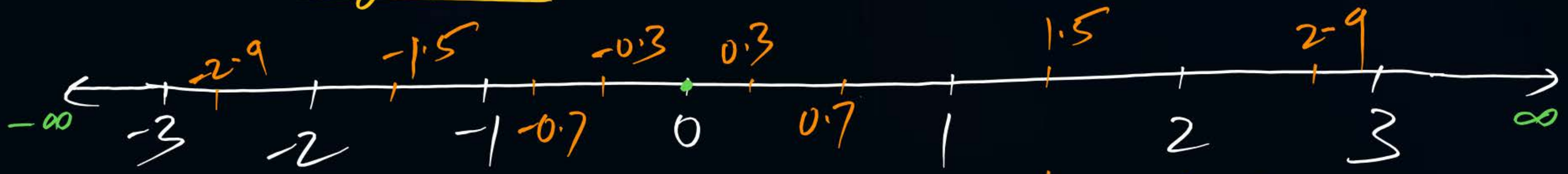


② If $y = |x| = \sqrt{x^2}$ then $\frac{dy}{dx} = \frac{d}{dx}|x| = \frac{|x|}{x}$ or $\frac{x}{|x|}$, $x \neq 0$

Proof: $\frac{d}{dx}|x| = \frac{d}{dx}\left(\sqrt{x^2}\right) = \frac{1}{2\sqrt{x^2}}\frac{d}{dx}(x^2) = \frac{1}{2\sqrt{x^2}}(2x) = \frac{x}{|x|} = \frac{x|x|}{|x|^2} = \frac{x|x|}{x^2} = \frac{|x|}{x}$, $x \neq 0$

i.e. $\frac{d}{dx}|x| = \frac{|x|}{x}, x \neq 0 = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \\ \text{DNE}, & x = 0 \end{cases} \neq \text{sgn}(x)$

(*) Greatest Integer func^n (Floor function) →



$\lfloor 0.3 \rfloor = 0$, $\lfloor 0.7 \rfloor = 0$, $\lfloor 1.5 \rfloor = 1$, $\lfloor 2.9 \rfloor = 2$ .....

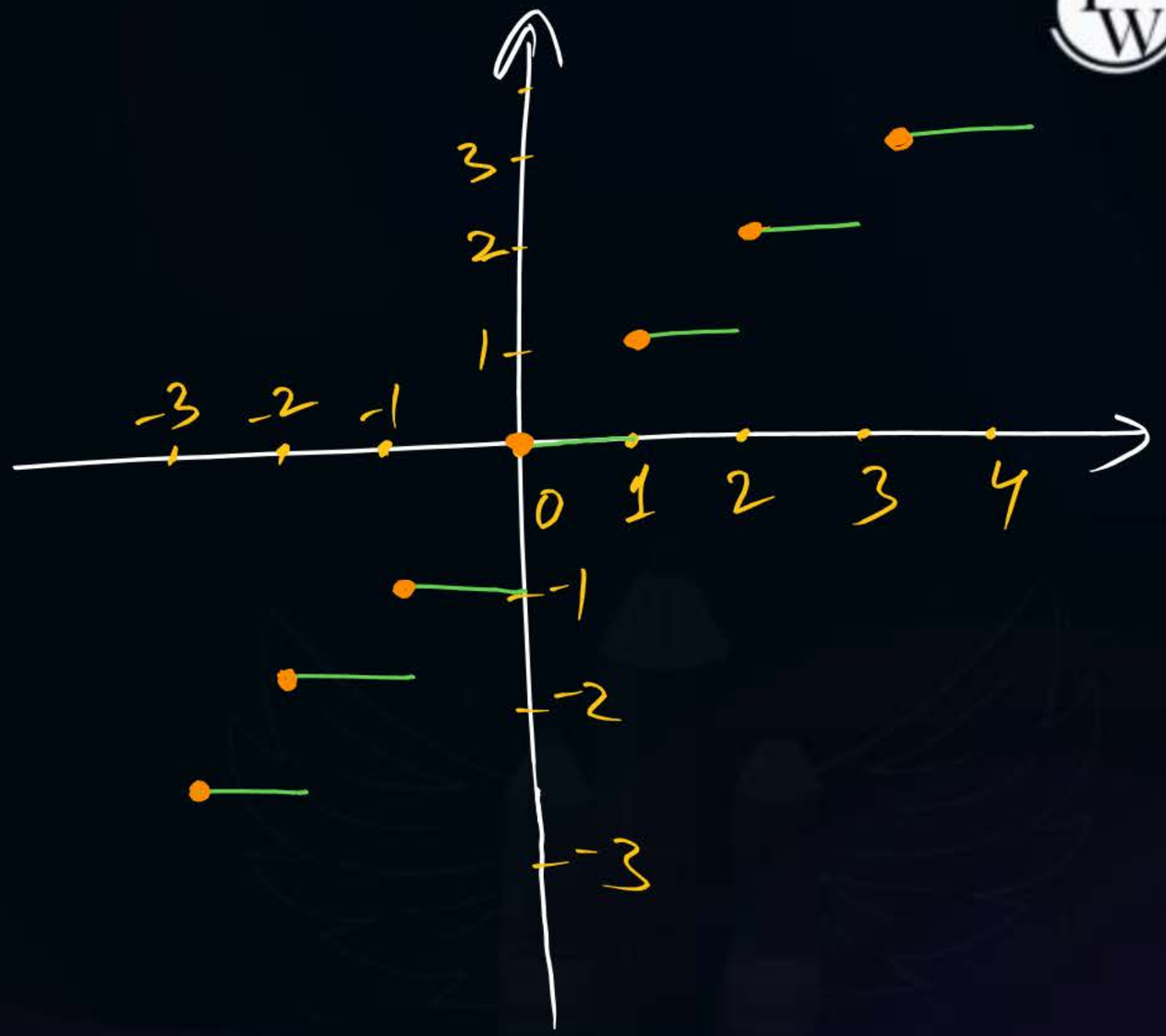
$\lfloor -0.3 \rfloor = -1$, $\lfloor -0.7 \rfloor = -1$, $\lfloor -1.5 \rfloor = -2$, $\lfloor -2.9 \rfloor = -3$ .....

$\lfloor 0 \rfloor = 0$, $\lfloor 1 \rfloor = 1$, $\lfloor 2 \rfloor = 2$, $\lfloor 3 \rfloor = 3$ .....

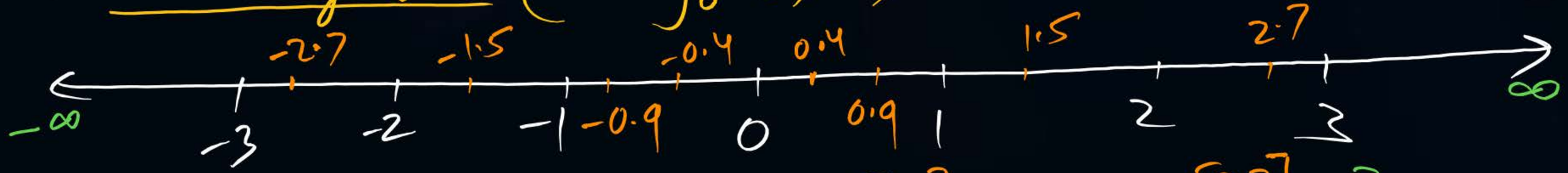
$\lfloor -1 \rfloor = -1$, $\lfloor -2 \rfloor = -2$, $\lfloor -3 \rfloor = -3$ .....

Graph of G.I.F → (Floor func^n)

$$y=[x]=\begin{cases} -3, & -3\le x<-2 \\ -2, & -2\le x<-1 \\ -1, & -1\le x<0 \\ 0, & 0\le x<1 \\ 1, & 1\le x<2 \\ 2, & 2\le x<3 \\ 3, & 3\le x<4 \\ \vdots & \end{cases}$$



⊛ Least Integer func^n (ceiling func^n) →



$\lceil 0.4 \rceil = 1$, $\lceil 0.9 \rceil = 1$, $\lceil 1.5 \rceil = 2$, $\lceil 2.7 \rceil = 3$ .....

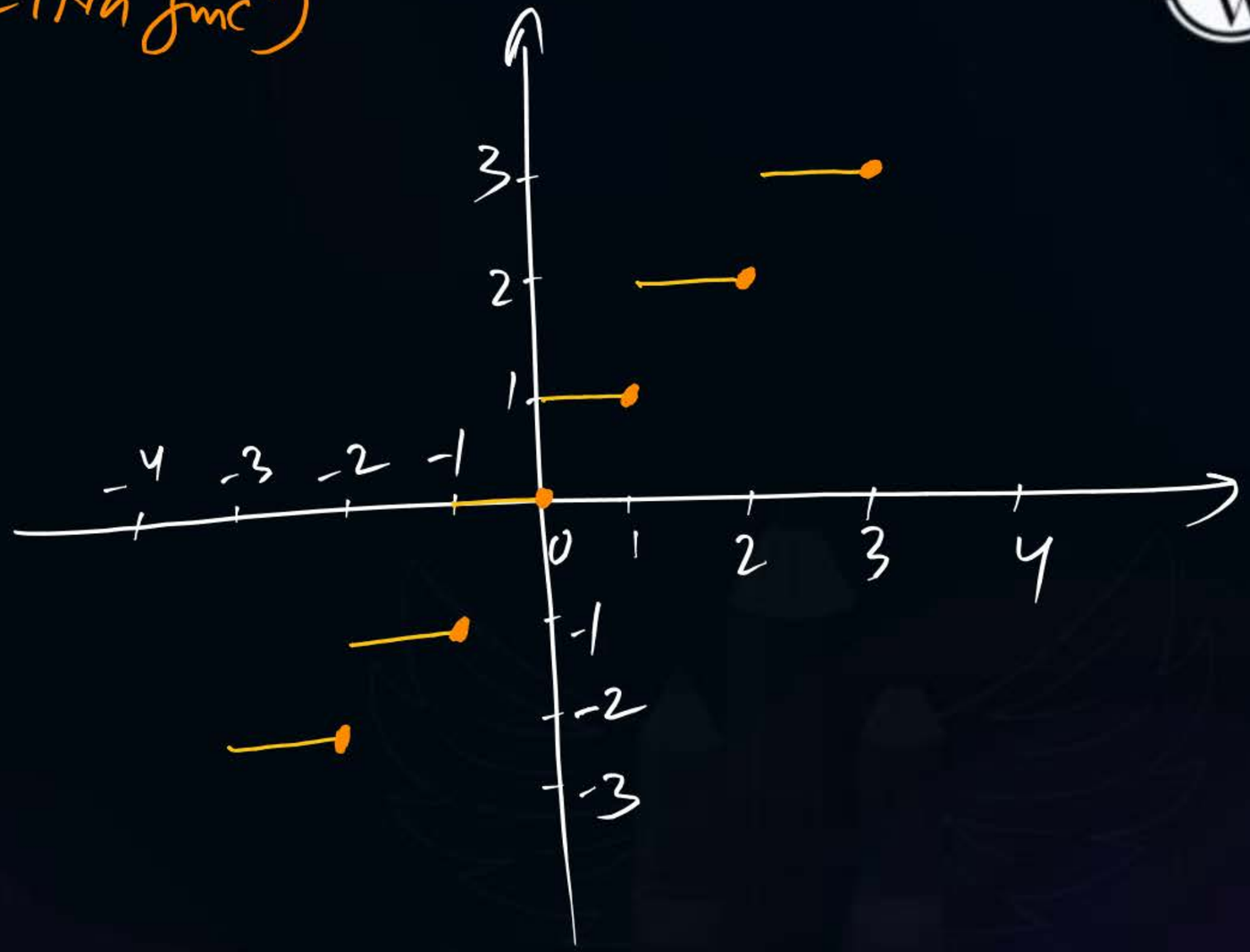
$\lceil -0.4 \rceil = 0$, $\lceil -0.9 \rceil = 0$, $\lceil -1.5 \rceil = -1$, $\lceil -2.7 \rceil = -2$ ....

$\lceil 0 \rceil = 0$, $\lceil 1 \rceil = 1$, $\lceil 2 \rceil = 2$, $\lceil 3 \rceil = 3$ .....

$\lceil -1 \rceil = -1$, $\lceil -2 \rceil = -2$, $\lceil -3 \rceil = -3$ .....

Least Integer function → (CEILING func^n)

$$y = \lceil x \rceil = \begin{cases} -1 & -2 < x \le -1 \\ 0 & -1 < x \le 0 \\ 1, & 0 < x \le 1 \\ 2, & 1 < x \le 2 \\ 3, & 2 < x \le 3 \\ 4 & 3 < x \le 4 \\ \vdots & \end{cases}$$



Fractional Part func^n : $y = \{x\} = \boxed{x - [x]}$


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THANK
you